

Real-Data Machine Learning for Rural Telemedicine Triage and Outbreak Surveillance

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Abstract—Rural telemedicine platforms face two simultaneous decisions: which individual patients can be managed remotely and which require urgent escalation, and whether population-level signals indicate a developing outbreak. We evaluate a data-science layer built exclusively on public, real-world data: the CDC NHAMCS 2019 emergency department survey (13,595 analysis encounters), the covid19india district-level archive (126,712 district-weeks across 840 districts), and a Kaggle pharmaceutical sales series (69 months). For triage, a class-weighted XGBoost model reaches cross-validated AUC 0.732 versus 0.719 for a logistic baseline. Operating-point selection reduces under-triage from 48.7% at the default threshold to 15.5% at $t = 0.25$ (recall 0.845), and a split-conformal calibration attains mean recall 0.850 over 200 randomized splits, with 46% of individual splits under the 0.85 target – a marginal rather than per-split guarantee. For surveillance, an IsolationForest operating at a fixed 5% alert budget enriches the Delta wave $7.0\times$ over its background flag rate with a median lead of 2 weeks; this is reported descriptively because no labeled outbreak ground truth exists. No synthetic data is used, and all code and seeds are released for exact reproduction.

Index Terms—machine learning, telemedicine, clinical triage, conformal prediction, outbreak surveillance, interpretable machine learning.

I. INTRODUCTION

Telemedicine services in resource-limited settings routinely carry two statistical workloads at once. At the patient level, an intake stream of voice transcripts and spot vital signs must be sorted into cases that a remote clinician can safely manage and cases that demand in-person escalation; missing an urgent case (under-triage) is a safety event, while over-triage consumes scarce clinician time. At the population level, the same platform sits on a growing archive of aggregated case counts, which can in principle provide early warning of local outbreaks before clinicians notice a pattern in individual visits. Both workloads are prediction problems, but they are severely differently supervised: individual dispositions are recorded, whereas district-level outbreak “truth” is not.

This paper evaluates the data-science layer of a rural telemedicine platform against that asymmetry, using only public real-world datasets. Our first module, patient-level triage, is built and benchmarked on the CDC NHAMCS 2019 emergency department (ED) survey [1], which carries vitals, reason-for-visit codes, and recorded triage acuity for 19,481 raw visits. Our second module, population-level surveillance, is built on the open covid19india district time-series archive [2]

covering 840 Indian districts. A supplementary module forecasts monthly pharmaceutical demand from a public Kaggle point-of-sale series (Zdravković [3]). Together they mirror the architecture of the deployed platform (Fig. 1) while keeping every evaluation number traceable to a committed artifact.

The paper makes four contributions:

- 1) A safety-first operating-point analysis for triage in which threshold selection (rather than model surgery) cuts the under-triage rate from 48.7% to 15.5% at recall 0.845, with an abstention band that routes low-confidence predictions to a physician.
- 2) An honest calibration study of split-conformal recall control: across 200 randomized splits the method attains mean recall 0.850 (SD 0.027), yet 46% of individual splits miss the 0.85 target, and we quantify exactly which guarantee holds.
- 3) A label-free outbreak evaluation design that scores detectors by empirical wave enrichment under a fixed alert budget rather than by invented precision/recall labels, showing $7.0\times$ Delta-wave concentration for single-week isolation flags.
- 4) A fully deterministic, fail-closed reproducibility pipeline: no synthetic data, no silently skipped steps, seed-42 reruns byte-identical, and every figure, table, and number regenerated from raw files by script.

Section II positions this against clinical scoring rules and conformal inference. Section III describes the data and protocols, Section IV reports results, and Section V states limitations plainly, including the US-to-India transfer gap.

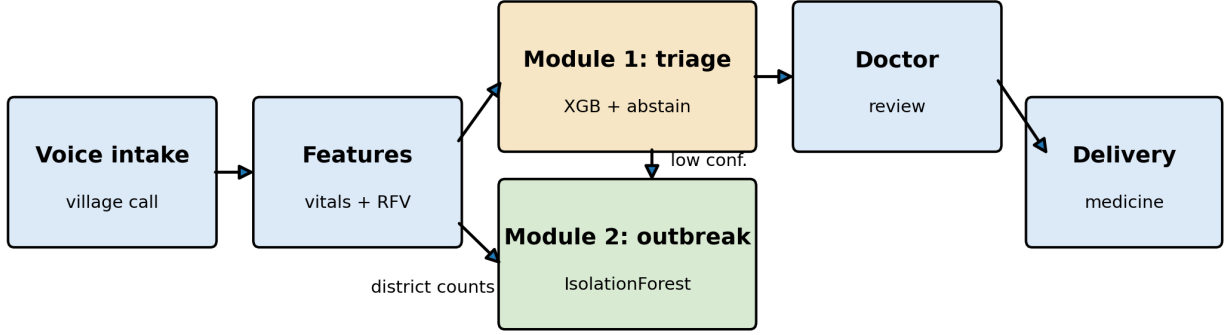
II. RELATED WORK

Clinical early-warning scores. The National Early Warning Score NEWS2 [4], [5] is the standard vitals-only acuity rule and the natural baseline for any triage model; we implement it directly as a comparator rather than assuming superiority over it.

Gradient boosting for clinical prediction. XGBoost [6] and LightGBM [7] are the de facto tabular learners, and scikit-learn [8] provides the linear baselines and splitting utilities used here. Our contribution is not a new learner but an operating-point and calibration treatment of an ordinary boosted model under an explicit asymmetric cost.

Distribution-free uncertainty. Conformal prediction [9] and its modern tutorials [10] give finite-sample coverage state-

Curago data-science layer: patient-level triage + population-level surveillance



Data: CDC NHAMCS 2019 (public domain) | covid19india archive (static) | Kaggle pharma sales (CC BY-NC, suppl.)

Fig. 1. The Curago data-science layer: voice intake and vitals with reason-for-visit features feed Module 1 (patient-level triage, XGBoost with an abstain band), while district counts feed Module 2 (population-level outbreak surveillance, IsolationForest); low-confidence cases route to a doctor, and forecasts support medicine delivery. Data: CDC NHAMCS 2019 (public domain); covid19india static archive; Kaggle pharma sales (CC BY-NC 4.0, supplementary).

ments under exchangeability. We use the split-conformal construction to control sensitivity rather than set size, and we report the finite-sample behavior across random splits because the textbook guarantee is marginal.

Interpretability. SHAP [11] attributions are used to audit what the triage model relies on, specifically whether it attends to clinically plausible vitals and reason-for-visit patterns.

Outbreak detection without labels. Frequency-threshold and scan-statistic detectors predate machine learning, and open dashboards such as OpenDengue [12] illustrate that curated public outbreak labels are rare. We therefore evaluate detectors descriptively against wave windows rather than against fabricated ground truth.

Demand forecasting. Seasonal-naïve and Holt-Winters [13] baselines are reported alongside LightGBM precisely because complex models do not automatically win on short monthly series.

III. DATA AND METHODS

A. Datasets

Triage (NHAMCS 2019 ED). The National Hospital Ambulatory Medical Care Survey 2019 emergency department file [1] is a US public-domain, de-identified survey. Restricting to records with a valid recorded triage level yields 13,595 analysis encounters from 19,481 raw records, with high-acuity prevalence 0.152. The target label marks encounters recorded as emergent or urgent (the escalation-worthy class under the platform’s routing policy). Features are age, sex, six vital signs (temperature, pulse, respiratory rate, systolic and diastolic pressure, oxygen saturation), pain scale, and dummy indicators for the 20 most frequent reason-for-visit codes decoded with official NCHS labels; missing vitals are median-imputed inside each training fold. Cohort medians and dataset extents are summarized in the supplementary cohort table (repository artifact `paper_T7_cohort`).

Outbreak (covid19india archive). The static district-level archive [2] spans 2020-04-26 to 2023-08-22 and is resampled to 126,712 district-weeks over 840 state-qualified districts. It contains reported case counts only: no outbreak labels exist at district-week resolution.

Supplementary demand (Kaggle pharma sales). A monthly point-of-sale series of eight ATC drug groups [3] (CC BY-NC 4.0) contributes 69 complete months; a partial collection month is excluded by rule, and the final 12 months form the holdout.

B. Triage model and operating rule

Two models are trained under a shared 5-fold stratified protocol (seed 42): a class-weighted logistic regression with standardization (the baseline) and a class-weighted XGBoost [6] (`n_estimators=200`, depth 6, learning rate 0.1, `scale_pos_weight` computed from training folds), both behind a median imputer using scikit-learn [8]. Under-triage is defined as $1 - \text{recall}$ on the high-acuity class.

Rather than accept the default 0.5 threshold, we select the operating point from the out-of-fold probability sweep as the lowest threshold whose recall reaches at least 0.80, tie-broken by precision; this yields $t = 0.25$ for XGBoost. A physician-abstention band routes any case with maximum class probability below 0.60 to human review regardless of threshold. Threshold tuning moves the model along its ROC curve; it does not change AUC.

C. Split-conformal recall control

To target a sensitivity level with a finite-sample statement, we wrap the trained classifier in split-conformal calibration [9], [10]: each replicate draws stratified 60/20/20 train/calibration/test splits; on the calibration set the decision threshold is set to the $\lfloor \alpha(n+1) \rfloor$ -th smallest nonconformity quantile among positives at error level $\alpha = 0.15$, i.e. a nominal

recall of $1 - \alpha = 0.85$. We repeat over 200 random seeds (0–199) and report the distribution of test recall, the implied thresholds, and abstention rates. The protocol uses train-fold statistics only. As a clinical reference we also evaluate NEWS2 [4], [5] computed from the vitals columns alone.

D. Outbreak surveillance under an alert budget

District-week counts are converted to weekly features and fed to an IsolationForest with contamination 0.05, which fixes the alert budget at 5% of district-weeks by construction; we never present that fraction as a detected-outbreak rate. Two variants (single-week flags and confirmation-filtered flags) are compared against a rolling 4-week z-score baseline [8]. Because no gold labels exist, evaluation is *descriptive enrichment*: for the Delta (2021-04-01–2021-06-30) and Omicron wave windows we report the flag rate inside the window versus the background flag rate outside it, district coverage, and median lead relative to window start, mirroring the alert-budget evaluation style of public health surveillance practice.

E. Reproducibility discipline

All pipelines fail closed: if a raw file is absent the script exits with the download URL instead of substituting synthetic data. Outputs are written as committed artifacts; seed-42 reruns reproduce every CSV and JSON byte-identically, and the paper’s numeric macros are generated from the committed `paper_numbers.json` rather than typed by hand.

IV. RESULTS

A. Triage discrimination and interpretability

Table I compares the two learners at the default threshold. XGBoost leads on AUC (0.732 vs 0.719) and precision, but at $t = 0.5$ it misses 0.487 of high-acuity patients, an unacceptable operating point for triage. Fig. 2 shows the ROC curves with the default, tuned, and conformal operating points marked: the entire useful operating region lies well below $t = 0.5$. SHAP attributions (supplementary figure, repository artifact `paper_fig2_shap`) confirm the model leans on age, pulse, respiratory rate, and pain score, with reason-for-visit codes contributing clinically recognizable splits such as chest pain and shortness of breath.

B. Operating-point selection

Table II and Fig. 3 show the central triage result: lowering the threshold from 0.50 to 0.25 cuts under-triage from 48.7% to 15.5% (recall 0.845) at the cost of precision (0.207) and abstention 0.235; over-triage rises correspondingly. This is the standard sensitivity-first trade of emergency triage, reported on both axes so the exchange is visible. The abstention band keeps 0.235 of cases with the model deferring to a physician, which bounds the blast radius of the precision drop.

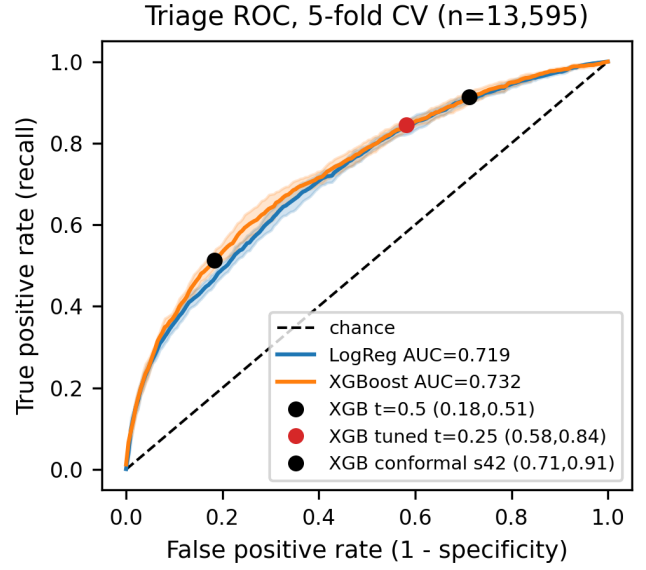


Fig. 2. Triage ROC under 5-fold cross-validation (13,595 analysis encounters): logistic regression AUC 0.719, XGBoost AUC 0.732. Marked operating points: default $t = 0.50$, tuned $t = 0.25$, and the seed-42 conformal split. Data: CDC NHAMCS 2019 ED.

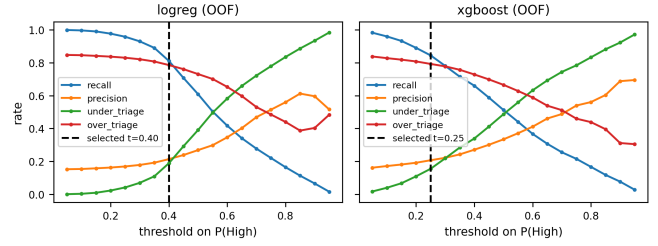


Fig. 3. Out-of-fold threshold trade-offs (recall, precision, under- and over-triage, abstention) with the pre-declared selections $t = 0.25$ (XGBoost). Data: CDC NHAMCS 2019 ED.

C. Conformal recall control

Table III sweeps the conformal error level over 200 splits per setting. At $\alpha = 0.15$ the mean test recall is 0.850 (SD 0.027), yet 46% of the splits miss the 0.85 mark (Fig. 4); the finite-sample statement is marginal over calibration draws, not a per-deployment promise, and we state it that way. Mean implied thresholds (0.212) and abstention (0.218) are stable across seeds.

On the seed-42 reference split the conformal selector reaches recall 0.913 (95% bootstrap CI 0.886–0.94), precision 0.187, AUC 0.739, Brier 0.158, abstention 0.231, at threshold 0.173. The vitals-only NEWS2 baseline cannot match the target on this cohort: no cutoff reaches the 0.85 recall level (cutoff ≥ 1 attains recall 0.669 at precision 0.168; the clinical cutoff ≥ 5 attains 0.138 at 0.207). The learned model therefore buys recall at comparable precision, but its reliability diagram (supplementary artifact `paper_figS1b_calibration`, Brier 0.158) shows overconfident probabilities: scores are rank-useful for the threshold rule and are not quoted as absolute

TABLE I
TRIAGE MODEL COMPARISON, 5-FOLD CROSS-VALIDATION (T=0.5, SEED 42).

Model	AUC	Recall	Precision	F1	Under-triage	Data Source
Logistic Regression (baseline)	0.719	0.609	0.269	0.373	0.391	CDC NHAMCS 2019 ED
XGBoost (class-weighted)	0.732	0.513	0.334	0.405	0.487	CDC NHAMCS 2019 ED

Data source: CDC NHAMCS 2019 ED (n=13,595).

TABLE II
OPERATING POINTS FOR THE TRIAGE MODULE: DEFAULT, TUNED (PRE-DECLARED RECALL ≥ 0.80 RULE) AND SPLIT-CONFORMAL (SEED 42).

Operating point	Threshold	Recall	Precision	Under-triage	Abstention	AUC	Data Source
XGBoost, t=0.50 (OOF)	0.5	0.513	0.334	0.487	0.235	0.732	CDC NHAMCS 2019 ED
XGBoost, t=0.25 tuned (OOF)	0.25	0.845	0.207	0.155	0.235	0.732	CDC NHAMCS 2019 ED
LogReg, t=0.40 tuned (OOF)	0.4	0.811	0.214	0.189	0.394	0.719	CDC NHAMCS 2019 ED
XGBoost, conformal, seed-42 test	0.173	0.913	0.187	0.087	0.231	0.739	CDC NHAMCS 2019 ED

Data source: CDC NHAMCS 2019 ED (n=13,595). AUC is threshold-independent (5-fold CV); OOF = out-of-fold.

TABLE III
SPLIT-CONFORMAL RECALL CONTROL OVER 200 RANDOM 60/20/20 SPLITS (XGBOOST, CALIBRATION QUANTILE ON POSITIVES). THE GUARANTEE IS MARGINAL: INDIVIDUAL SPLITS VARY.

alpha (miss target)	Mean recall	SD	Min	Max	Frac. splits ≥ 0.85	Mean precision	Data Source
0.05	0.953	0.015	0.911	0.983	0.0	0.171	CDC NHAMCS 2019 ED (n=13,595)
0.1	0.902	0.02	0.841	0.954	0.005	0.185	CDC NHAMCS 2019 ED (n=13,595)
0.15	0.85	0.027	0.773	0.92	0.46	0.198	CDC NHAMCS 2019 ED (n=13,595)
0.2	0.8	0.029	0.729	0.877	0.97	0.211	CDC NHAMCS 2019 ED (n=13,595)
0.3	0.7	0.032	0.604	0.795	1.0	0.241	CDC NHAMCS 2019 ED (n=13,595)

Data source: CDC NHAMCS 2019 ED (n=13,595). Alpha 0.15 is the pre-declared main setting (also conformal_repeats.csv, 200 seeds).

Conformal recall over 200 splits (marginal guarantee)

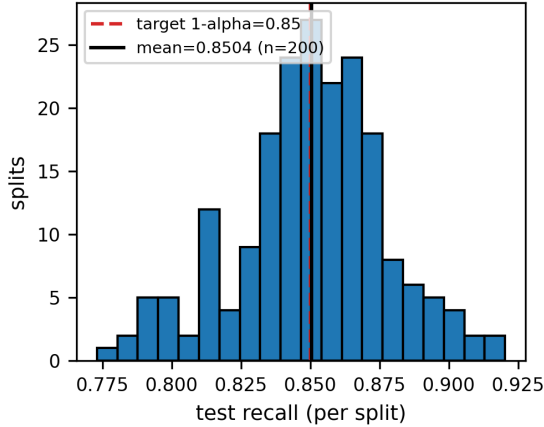


Fig. 4. Split-conformal test recall over 200 seeds at $\alpha = 0.15$; mean 0.850 vs the 0.85 target; 46% of individual splits fall below target. The guarantee is marginal, not per-split. Data: CDC NHAMCS 2019 ED.

risks.

D. Outbreak surveillance

Table IV and Fig. 5 compare detectors descriptively. Single-week IsolationForest flags concentrate in the Delta window at rate 0.25 versus 0.036 outside, an enrichment of 7.0 $\times$, with district coverage 0.398 and median lead 2 weeks over a background flag rate of 0.024. Requiring a confirmation week

Outbreak methods vs epidemic windows (descriptive)

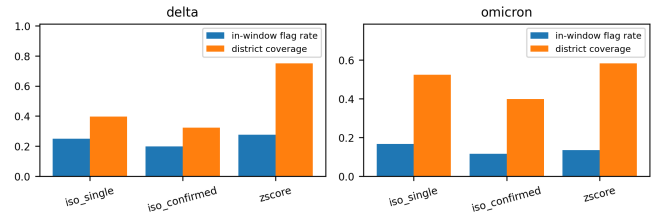


Fig. 5. Outbreak methods versus the Delta and Omicron windows (in-window flag rate and district coverage): descriptive enrichment only; single-week isolation flags enrich Delta 7.0 $\times$. Data: covid19india static archive, 840 districts.

halves the background to 0.013 but costs a week of lead; the z-score baseline leads earliest and covers the widest share of districts, at roughly four times the background flag rate of single-week isolation flags (0.093). None of these numbers is a precision or recall: without district-level outbreak labels, honest claims stop at enrichment.

E. Supplementary demand forecast

On the 12-month holdout over 69 months, per-group MAE winners are Holt-Winters on 4 of 8 drug groups, seasonal naive on 2, and LightGBM on 2 (supplementary forecast table, repository artifact `paper_TS1_forecast`). The honest reading is that classical baselines remain competitive on short monthly series; this module supports logistics, not clinical

TABLE IV

OUTBREAK DETECTION METHODS VERSUS PRE-DECLARED EPIDEMIC WINDOWS (DESCRIPTIVE; NO LABELED GROUND TRUTH EXISTS). BACKGROUND RATE = FLAG SHARE OF DISTRICT-WEEKS OUTSIDE BOTH WINDOWS. ISOLATIONFOREST SINGLE-WEEK FLAGS ENRICH THE DELTA WINDOW $7.0\times$ (0.250 INSIDE VS 0.036 OUTSIDE).

Method	Window	In-window flag rate	District coverage	Median lead (weeks)	Background rate	Data Source
iso_single	Delta	0.25	0.398	2.0	0.024	covid19india static archive
iso_single	Omicron	0.167	0.525	2.0	0.024	covid19india static archive
iso_confirmed	Delta	0.199	0.324	1.0	0.013	covid19india static archive
iso_confirmed	Omicron	0.116	0.399	1.0	0.013	covid19india static archive
zscore	Delta	0.276	0.751	4.0	0.093	covid19india static archive
zscore	Omicron	0.135	0.583	2.0	0.093	covid19india static archive

Data source: covid19india static archive (data.incovid19.org, retrieved 2026-10-07); 840 state-qualified districts, 126,712 district-weeks.

claims, and is excluded from the main findings (supplementary figures in the repository).

V. DISCUSSION

a) What holds.: On this cohort, a standard boosted model with a deliberately chosen threshold and a physician-abstention band delivers a large, auditable reduction in under-triage, and split-conformal calibration provides a workable mechanism for targeting sensitivity with a stated error level. The surveillance module shows that even a label-free detector concentrates alerts in known wave windows far above its background rate, which is the strongest claim the data support.

b) Limitations.: *Transfer.* NHAMCS is US ED data; rural-Indian intake populations, disease mix, and vital-sign distributions differ. Conformal coverage holds under exchangeability within this sample and does not automatically transfer under geographic covariate shift; in deployment the model is a decision-support prior behind abstention and physician verification. *Guarantee strength.* The conformal guarantee is marginal: 46% of individual splits miss the target, so a single unlucky calibration split can under-deliver. *Calibration.* Tree probabilities are overconfident (Brier 0.158); they are used for ranking and thresholds, not as quoted risks. *Labels.* Outbreak evaluation is descriptive enrichment against wave windows, not precision/recall; the archive is retrospective, not a live surveillance feed. *Protocol.* Triage numbers combine 5-fold cross-validation with a single-protocol split study; there is no second external test set, and the demand forecast uses a single holdout without rolling origin evaluation.

VI. DATA AVAILABILITY, LICENSING, AND ETHICS

All three datasets are public: NHAMCS 2019 ED is US public domain from the CDC/NCHS [1]; the covid19india district archive is an open static release [2]; the Kaggle pharma-sales series is CC BY-NC 4.0 [3]. No private, proprietary, or synthetic patient data was used. The analysis uses de-identified, aggregated, or public survey records, so no institutional approval or individual consent applies; license terms are restated in the repository’s data manifest. Code, ingestion, training, and figure/table generation scripts are released under the MIT License at https://github.com/vaibhav7087/curago_paper_version, and every reported number is regenerated from raw files by a deterministic, fail-closed pipeline.

VII. CONCLUSION

Pairing a threshold-tuned, abstention-guarded triage model with a label-free outbreak detector turns a telemedicine platform’s raw streams into two quantified decisions: which patients escalate, and where to look next. The results are deliberately bounded—48.7% under-triage reduced to 15.5% at recall 0.845, a marginal conformal guarantee of 0.850 mean recall, and $7.0\times$ descriptive wave enrichment—and every number is reproducible from public data with released code. Future work is prospective validation on rural-Indian intake data and calibration monitoring under drift.

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